

Krish Jeswal

Full-Stack Developer & ML Scientist | Electronics & Telecom Undergrad

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Summary

Versatile engineer with a proven track record of shipping real products, from production React/TypeScript web apps with serverless backends to ML pipelines and embedded systems using ESP32 and Python. Active in hackathons, CTFs, and developer communities, with experience leading dev teams. Comfortable across the full stack, ML/data workflows, and the hardware-software boundary, bringing both systems thinking and fast product delivery to whatever I build.

Education

R.V. College of Engineering Electronics and Telecommunication	Undergraduate Bengaluru, India • 2024 - 2028
Amity International School Physics, Chemistry, Maths and Computer Science	Higher Secondary Ghaziabad, India • 2009 - 2023

Skills

Programming Languages Python, C, JavaScript, TypeScript, Verilog	Frontend React, Tailwind CSS, Vite, Three.js, Framer Motion	Backend Node.js, EXPRESS, REST API, Flask, Docker
Databases & Tools PostgreSQL, Supabase, Firebase, Git, Google Cloud Platform	ML/Data NumPy, Pandas, sci-kit learn, OpenCV, Streamlit, Vertex AI	Electronics Arduino, ESP32, FPGA, UART, LTSpice, KiCAD

Projects

Synaptic-BioMemristor

- Designed an end-to-end hardware-software platform using **ESP32 + INA219** to characterize a bio-ionic memristor through voltage pulse/sweep studies, measuring I-V hysteresis, ON/OFF ratio, retention, and synaptic behavior.
- Trained a **Linear Regression model** with **scikit-learn** on experimentally collected data to predict future conductance states of the device.

Intelligent Excel Parser

- Built a **FastAPI** service using **Google Gemini (gemini-2.5-flash-lite)** to intelligently map inconsistent .xlsx column headers to a canonical registry, with automatic fallback to **SequenceMatcher** similarity scoring.
- Engineered a modular pipeline with **openpyxl**, **Pydantic v2** schema validation, and multi-format value parsing; returns structured **JSON** with per-cell confidence levels via an interactive web dashboard.

MoodVue

- Extended a real-time emotion detection app, integrating **DeepFace + OpenCV** via an async **Flask** backend to analyze live webcam feeds and track mood patterns over time.
- Built mood history visualization with **Recharts** in **React 18 + TypeScript**; integrated **Supabase** for user authentication and data persistence; containerized with **Docker** and deployed on **Vercel + Render**.

Experience

Google Developer Groups - Research & Projects Completed Google Cloud Study Jams, building practical skills across core GCP services including compute, storage, and ML tooling. Co-volunteered at the Vision 2047 Hackathon, supporting event coordination and engaging with teams working on socially impactful tech solutions.	Jun 2025 – Present
Coding Club, RVCE - Member Strengthened CS fundamentals through regular collaborative problem-solving sessions focused on algorithms, data structures, and competitive programming. Actively participated in and volunteered at multiple hackathons and CTFs, gaining hands-on experience with time-constrained problem solving and team coordination.	Dec 2024 – Present
The Institution of Electronics and Telecommunication Engineers (IETE) - Member Led the development team for the official Student Forum website, overseeing architecture decisions, task delegation, and end-to-end delivery. Contributed to broader technical initiatives within the chapter, including hackathon-style projects and cross-team collaborations on engineering challenges.	Nov 2025 – Present